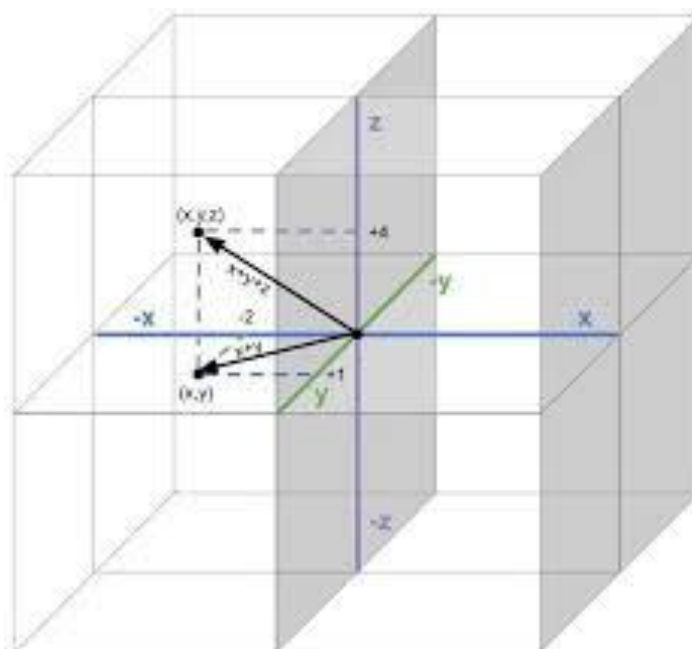


Math 1057

Instructor

Dr. Mortaza Baký Haskuee



21 Eigenvalues and Eigenvectors

21.1 Eigenvectors and Eigenvalues 163

21.2 When $\lambda = 0$ is an eigenvalue 168

Lecture 21

Eigenvalues and Eigenvectors

21.1 Eigenvectors and Eigenvalues

An $n \times n$ matrix $\mathbf{A}$ can be thought of as the linear mapping that takes any arbitrary vector $\mathbf{x} \in \mathbb{R}^n$ and outputs a new vector $\mathbf{Ax}$. In some cases, the new output vector $\mathbf{Ax}$ is simply a scalar multiple of the input vector $\mathbf{x}$, that is, there exists a scalar λ such that $\mathbf{Ax} = \lambda\mathbf{x}$. This case is so important that we make the following definition.

Definition 21.1: Let $\mathbf{A}$ be a $n \times n$ matrix and let $\mathbf{v}$ be a non-zero vector. If $\mathbf{Av} = \lambda\mathbf{v}$ for some scalar λ then we call the vector $\mathbf{v}$ an **eigenvector** of $\mathbf{A}$ and we call the scalar λ an **eigenvalue** of $\mathbf{A}$ corresponding to $\mathbf{v}$.

Hence, an eigenvector $\mathbf{v}$ of $\mathbf{A}$ is simply scaled by a scalar λ under multiplication by $\mathbf{A}$. Eigenvectors are by definition nonzero vectors because $\mathbf{A}\mathbf{0}$ is clearly a scalar multiple of $\mathbf{0}$ and then it is not clear what that the corresponding eigenvalue should be.

Example 21.2. Determine if the given vectors $\mathbf{v}$ and $\mathbf{u}$ are eigenvectors of $\mathbf{A}$? If yes, find the eigenvalue of $\mathbf{A}$ associated to the eigenvector.

$$\mathbf{A} = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

Solution. Compute

$$\begin{aligned} \mathbf{Av} &= \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \\ 2 \end{bmatrix} \\ &= 2 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \\ &= 2\mathbf{v} \end{aligned}$$

Hence, $\mathbf{A}\mathbf{v} = 2\mathbf{v}$ and thus $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with corresponding eigenvalue $\lambda = 2$. On the other hand,

$$\mathbf{A}\mathbf{u} = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ 4 \end{bmatrix}.$$

There is no scalar λ such that

$$\begin{bmatrix} 0 \\ 6 \\ 4 \end{bmatrix} = \lambda \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

Therefore, $\mathbf{u}$ is not an eigenvector of $\mathbf{A}$. □

Example 21.3. Is $\mathbf{v}$ an eigenvector of $\mathbf{A}$? If yes, find the eigenvalue of $\mathbf{A}$ associated to $\mathbf{v}$:

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -4 & 2 & 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Solution. We compute

$$\mathbf{A}\mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}.$$

Hence, if $\lambda = 0$ then $\lambda\mathbf{v} = \mathbf{0}$ and thus $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. Therefore, $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with corresponding eigenvalue $\lambda = 0$. □

How does one find the eigenvectors/eigenvalues of a matrix $\mathbf{A}$? The general procedure is to first find the eigenvalues of $\mathbf{A}$ and then for each eigenvalue find the corresponding eigenvectors. In this section, however, we will instead suppose that we have already found the eigenvalues of $\mathbf{A}$ and concern ourselves with finding the associated eigenvectors. Suppose then that λ is known to be an eigenvalue of $\mathbf{A}$. How do we find an eigenvector $\mathbf{v}$ corresponding to the eigenvalue λ ? To answer this question, we note that if $\mathbf{v}$ is to be an eigenvector of $\mathbf{A}$ with eigenvalue λ then $\mathbf{v}$ must satisfy the equation

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

We can rewrite this equation as

$$\mathbf{A}\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$$

which, after using the distributive property of matrix multiplication, is equivalent to

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}.$$

The last equation says that if $\mathbf{v}$ is to be an eigenvector of $\mathbf{A}$ with eigenvalue λ then $\mathbf{v}$ must be in the null space of $\mathbf{A} - \lambda\mathbf{I}$:

$$\mathbf{v} \in \text{Null}(\mathbf{A} - \lambda\mathbf{I}).$$

In summary, if λ is known to be an eigenvalue of $\mathbf{A}$, then to find the eigenvectors corresponding to λ we must solve the homogeneous system

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}.$$

Recall that the null space of any matrix is a subspace and for this reason we call the subspace $\text{Null}(\mathbf{A} - \lambda\mathbf{I})$ the **eigenspace** of $\mathbf{A}$ corresponding to λ .

Example 21.4. It is known that $\lambda = 4$ is an eigenvalue of

$$\mathbf{A} = \begin{bmatrix} -4 & 6 & 3 \\ 1 & 7 & 9 \\ 8 & -6 & 1 \end{bmatrix}.$$

Find a basis for the eigenspace of $\mathbf{A}$ corresponding to $\lambda = 4$.

Solution. First compute

$$\mathbf{A} - 4\mathbf{I} = \begin{bmatrix} -4 & 6 & 3 \\ 1 & 7 & 9 \\ 8 & -6 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} -8 & 6 & 3 \\ 1 & 3 & 9 \\ 8 & -6 & -3 \end{bmatrix}$$

Find a basis for the null space of $\mathbf{A} - 4\mathbf{I}$:

$$\begin{aligned} \begin{bmatrix} -8 & 6 & 3 \\ 1 & 3 & 9 \\ 8 & -6 & -3 \end{bmatrix} &\xrightarrow{R_1 \uparrow R_2} \begin{bmatrix} 1 & 3 & 9 \\ -8 & 6 & 3 \\ 8 & -6 & -3 \end{bmatrix} \\ \begin{bmatrix} 1 & 3 & 9 \\ -8 & 6 & 3 \\ 8 & -6 & -3 \end{bmatrix} &\xrightarrow{\substack{8R_1 + R_2 \\ -8R_1 + R_3}} \begin{bmatrix} 1 & 3 & 9 \\ 0 & 30 & 75 \\ 0 & -30 & -75 \end{bmatrix} \end{aligned}$$

Finally,

$$\begin{bmatrix} 1 & 3 & 9 \\ 0 & 30 & 75 \\ 0 & -30 & -75 \end{bmatrix} \xrightarrow{R_2 + R_3} \begin{bmatrix} 1 & 3 & 9 \\ 0 & 30 & 75 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, the general solution to the homogenous system $(\mathbf{A} - 4\mathbf{I})\mathbf{x} = \mathbf{0}$ is

$$\mathbf{x} = t \begin{bmatrix} -3/2 \\ -5/2 \\ 1 \end{bmatrix}$$

where t is an arbitrary scalar. Therefore, the eigenspace of $\mathbf{A}$ corresponding to $\lambda = 4$ is

$$\text{span} \left\{ \begin{bmatrix} -3/2 \\ -5/2 \\ 1 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} -3 \\ -5 \\ 2 \end{bmatrix} \right\} = \text{span}\{\mathbf{v}\}$$

and $\{\mathbf{v}\}$ is a basis for the eigenspace. The vector $\mathbf{v}$ is of course an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda = 4$ and also (of course) any multiple of $\mathbf{v}$ is also eigenvector of $\mathbf{A}$ with $\lambda = 4$. $\square$

Example 21.5. It is known that $\lambda = 3$ is an eigenvalue of

$$\mathbf{A} = \begin{bmatrix} 11 & -4 & -8 \\ 4 & 1 & -4 \\ 8 & -4 & -5 \end{bmatrix}.$$

Find the eigenspace of $\mathbf{A}$ corresponding to $\lambda = 3$.

Solution. First compute

$$\mathbf{A} - 3\mathbf{I} = \begin{bmatrix} 11 & -4 & -8 \\ 4 & 1 & -4 \\ 8 & -4 & -5 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 8 & -4 & -8 \\ 4 & -2 & -4 \\ 8 & -4 & -8 \end{bmatrix}$$

Now find the null space of $\mathbf{A} - 3\mathbf{I}$:

$$\begin{aligned} \begin{bmatrix} 8 & -4 & -8 \\ 4 & -2 & -4 \\ 8 & -4 & -8 \end{bmatrix} &\xrightarrow{R_1 \uparrow R_2} \begin{bmatrix} 4 & -2 & -4 \\ 8 & -4 & -8 \\ 8 & -4 & -8 \end{bmatrix} \\ \begin{bmatrix} 4 & -2 & -4 \\ 8 & -4 & -8 \\ 8 & -4 & -8 \end{bmatrix} &\xrightarrow{\substack{-2R_1+R_2 \\ -2R_1+R_3}} \begin{bmatrix} 4 & -2 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Hence, any vector in the null space of

$$\mathbf{A} - 3\mathbf{I} = \begin{bmatrix} 4 & -2 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

can be written as

$$\mathbf{x} = t_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t_2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Therefore, the eigenspace of $\mathbf{A}$ corresponding to $\lambda = 3$ is

$$\text{Null}(\mathbf{A} - 3\mathbf{I}) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \right\}.$$

The vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are two linearly independent eigenvectors of $\mathbf{A}$ with eigenvalue $\lambda = 3$. Therefore $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for the eigenspace of $\mathbf{A}$ with eigenvalue $\lambda = 3$. You can verify that $\mathbf{A}\mathbf{v}_1 = 3\mathbf{v}_1$ and $\mathbf{A}\mathbf{v}_2 = 3\mathbf{v}_2$. $\square$

As shown in the last example, there may exist more than one linearly independent eigenvector of $\mathbf{A}$ corresponding to the same eigenvalue, in other words, it is possible that the dimension of the eigenspace $\text{Null}(\mathbf{A} - \lambda\mathbf{I})$ is greater than one. What can be said about the eigenvectors of $\mathbf{A}$ corresponding to different eigenvalues?

Theorem 21.6: Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be eigenvectors of $\mathbf{A}$ corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_k$ of $\mathbf{A}$. Then $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a linearly independent set.

Solution. Suppose by contradiction that $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly dependent and $\{\lambda_1, \dots, \lambda_k\}$ are distinct. Then, one of the eigenvectors $\mathbf{v}_{p+1}$ that is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_p$, and $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly independent:

$$\mathbf{v}_{p+1} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p. \quad (21.1)$$

Applying $\mathbf{A}$ to both sides we obtain

$$\mathbf{A}\mathbf{v}_{p+1} = c_1\mathbf{A}\mathbf{v}_1 + c_2\mathbf{A}\mathbf{v}_2 + \dots + c_p\mathbf{A}\mathbf{v}_p$$

and since $\mathbf{A}\mathbf{v}_i = \lambda_i\mathbf{v}_i$ we can simplify this to

$$\lambda_{p+1}\mathbf{v}_{p+1} = c_1\lambda_1\mathbf{v}_1 + c_2\lambda_2\mathbf{v}_2 + \dots + c_p\lambda_p\mathbf{v}_p. \quad (21.2)$$

On the other hand, multiply (21.1) by λ_{p+1} :

$$\lambda_{p+1}\mathbf{v}_{p+1} = c_1\lambda_{p+1}\mathbf{v}_1 + c_2\lambda_{p+1}\mathbf{v}_2 + \dots + c_p\lambda_{p+1}\mathbf{v}_p. \quad (21.3)$$

Now subtract equations (21.2) and (21.3):

$$\mathbf{0} = c_1(\lambda_1 - \lambda_{p+1})\mathbf{v}_1 + c_2(\lambda_2 - \lambda_{p+1})\mathbf{v}_2 + \dots + c_p(\lambda_p - \lambda_{p+1})\mathbf{v}_p.$$

Now $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly independent and thus $c_i(\lambda_i - \lambda_{p+1}) = 0$. But the eigenvalues $\{\lambda_1, \dots, \lambda_k\}$ are all distinct and so we must have $c_1 = c_2 = \dots = c_p = 0$. But from (21.1) this implies that $\mathbf{v}_{p+1} = \mathbf{0}$, which is a contradiction because eigenvectors are by definition non-zero. This proves that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a linearly independent set. $\square$

Example 21.7. It is known that $\lambda_1 = 1$ and $\lambda_2 = -1$ are eigenvalues of

$$\mathbf{A} = \begin{bmatrix} -4 & 6 & 3 \\ 1 & 7 & 9 \\ 8 & -6 & 1 \end{bmatrix}.$$

Find bases for the eigenspaces corresponding to λ_1 and λ_2 and show that any two vectors from these distinct eigenspaces are linearly independent.

Solution. Compute

$$\mathbf{A} - \lambda_1\mathbf{I} = \begin{bmatrix} -5 & 6 & 3 \\ 1 & 6 & 9 \\ 8 & -6 & 0 \end{bmatrix}$$

and one finds that

$$(\mathbf{A} - \lambda_1\mathbf{I}) = \text{span} \left\{ \begin{bmatrix} -3 \\ -4 \\ 3 \end{bmatrix} \right\}$$

Hence, $\mathbf{v}_1 = (-3, -4, 3)$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda_1 = 1$, and $\{\mathbf{v}_1\}$ forms a basis for the corresponding eigenspace. Next, compute

$$\mathbf{A} - \lambda_2 \mathbf{I} = \begin{bmatrix} -4 & 6 & 3 \\ 1 & 7 & 9 \\ 8 & -6 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 6 & 3 \\ 1 & 8 & 9 \\ 8 & -6 & 2 \end{bmatrix}$$

and one finds that

$$\mathbf{A} - \lambda_2 \mathbf{I} = \text{span} \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Hence, $\mathbf{v}_2 = (-1, -1, 1)$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda_2 = -1$, and $\{\mathbf{v}_2\}$ forms a basis for the corresponding eigenspace. Now verify that $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent:

$$[\mathbf{v}_1 \quad \mathbf{v}_2] = \begin{bmatrix} -3 & -1 \\ -4 & -1 \\ 3 & 1 \end{bmatrix} \xrightarrow{R_1+R_3} \begin{bmatrix} -3 & -1 \\ -4 & -1 \\ 0 & 0 \end{bmatrix}$$

The last matrix has rank $r = 2$, and thus $\mathbf{v}_1, \mathbf{v}_2$ are indeed linearly independent. □

21.2 When $\lambda = 0$ is an eigenvalue

What can we say about $\mathbf{A}$ if $\lambda = 0$ is an eigenvalue of $\mathbf{A}$? Suppose then that $\mathbf{A}$ has eigenvalue $\lambda = 0$. Then by definition, there exists a non-zero vector $\mathbf{v}$ such that

$$\mathbf{A}\mathbf{v} = 0 \cdot \mathbf{v} = \mathbf{0}.$$

In other words, $\mathbf{v}$ is in the null space of $\mathbf{A}$. Thus, $\mathbf{A}$ is not invertible (Why?).

Theorem 21.8: The matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible if and only if $\lambda = 0$ is not an eigenvalue of $\mathbf{A}$.

In fact, later we will see that $\det(\mathbf{A})$ is the product of its eigenvalues.

After this lecture you should know the following:

- what eigenvalues are
- what eigenvectors are and how to find them when eigenvalues are known
- the behavior of a discrete dynamical system when the initial condition is set to an eigenvector of the system matrix